

Essentials of Digital Marketing

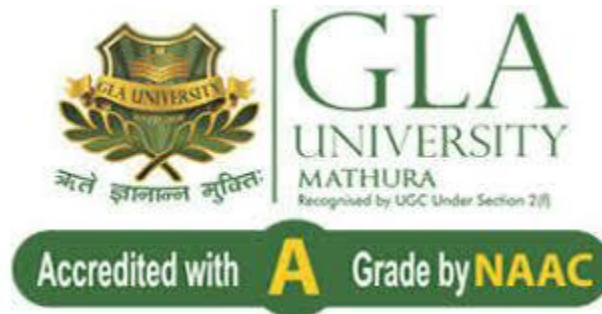
Module III

Mobile Marketing
Marketing Analytics & ROI

3.2 Marketing Analytics & ROI

- Key marketing engagement & ROI metrics,
- Primer on data science and analytics concepts,
- Web Traffic nuances,
- Multi-channel Analytics,
- Decoding CLV and RFM,
- Deciphering conversion and goal metrics,
- Implement conjoint analysis & decision tree tactics,
- Avoiding common analytical pitfalls.

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3.2 Marketing Analytics and ROI

Interesting Facts

- People on an average nowadays, spend more time online than on TV and other media (newspapers, magazines, etc.) combined.
- Total spending on Internet advertising is predicted to grow 12.9% next year.
- The Internet will become the largest medium for advertising in 2019 (ahead of TV).
- Less than 30% of small businesses use Web site analytics.
- 18% of small businesses admit that they do not tracking anything.
- Users who are retargeted to, are more likely to convert by 70%.

Analytics and ROI – The Connection

- **Marketing analytics** is the practice of
 - measuring,
 - managing and
 - analyzingmarketing performance to maximize its effectiveness and optimize return on investment (ROI).
- It allows to monitor campaigns and their respective outcomes, enabling companies to spend each dollar as effectively as possible.

- Beyond the obvious sales and lead generation applications, marketing analytics can offer profound insights into **customer preferences** and **trends**.
- Despite these compelling benefits, a majority of organizations fail to ever realize the promises of marketing analytics.
- According to a survey of senior marketing executives published in the Harvard Business Review,

"more than 80% of respondents were dissatisfied with their ability to measure marketing ROI."

The Need of Marketing Analytics

- A simple question with even simpler answer - **“To get the most out of your available resources”**.
- Marketing analytics helps to re-structure unorganized data into valuable insights using attribution.
- It puts the data in a visualization mode, revealing customer journeys across devices, geographies and more.

Need and importance of Marketing Analytics

- **Critical in making marketing budget**
 - Marketing is a significant expense and leaders want to know exactly what they're getting for it.
 - Marketing analytics helps in setting up a detailed roadmap that outlines the strategies, costs and projected results over a period.
 - It's also a great opportunity to focus on the future, generate new ideas, and inspire the team

- **Deciding where to spend**

- Marketers often calculate ROI at the channel or campaign level to determine **which efforts have a higher return** and therefore deserve additional investment.
- When **experimenting with a new ad network or marketing channel**, a crucial first step for marketers is **setting up analytics** such that marketing teams can measure ROI and determine if the network or channel is driving performance.
- Thus, a critical piece of the decision making, **to spend wisely and on right channels** can be driven by analytics.

- **Smart Optimization**

- Marketers often **optimize ongoing campaigns on the fly** to maximize performance based on the analytics reports.
- Effective optimizations require **digging deeper into spend and performance data**, to achieve more granular levels of reporting.
- Marketers might also seek to inform their optimizations with ROI analysis at the **Campaign, Creative, Keyword, Geographic or User level** — or any combination of the aforementioned dimensions.

3.1.1 Key Marketing Engagement & ROI metrics

Brand Awareness:

- This metric is used to show brand awareness.
 - It is evaluated based on the traffic driven to website,
 - how much people share the content,
 - the value of mentions and the reach of social messaging.
- Learn about brand awareness from **survey findings** and the amount of **news coverage over time**.
- Brands must measure it against **some kind of baseline**, which may include the historical data captured year over year.

Lead Generation

- This metric identifies **potential customers who may be interested in the products or services** being offered.
- Many companies use **dedicated landing pages on their websites** to capture information from these leads, for example: an email address, mailing address or phone numbers.
- Brands may want to deliver metrics on the average number of **leads generated weekly or monthly**.
- Some companies use **lead generation around specific events**, both online and in person, such as a **webinar or a trade show** to capture this metric.
- Lastly, brands may also **track the percentage of leads brought in** over a period versus the percentage of those **that converted to a sale**.

Customer Acquisition:

- **Bringing in new customers** is one of the best ways to show the success of marketing campaigns.
- However, every customer comes with the **cost of acquisition**.
- Add up **all the marketing expenses** and then divide this by **number of customers brought** into the business over a period.
- Other customer acquisition metrics include the **time it takes to close the sale** and the **average sale value**, which is sales profit divided by the number of sales converted.

Customer Loyalty:

- One of the other big goals is **to retain the customer**.
- This metric is used to show the **number of loyal customers** and also includes the **retention rate of the customers**.

Lifetime Value:

- It is a measure that predicts customer spending.
- How much **gross profit** can be attributed **to a particular customer** for the duration or the future of relationship.
- While measuring the campaign goals, brands must make sure that they **track metrics over time** and always **benchmark them against past performance or competitors**- if the data is available.

Case Study: Revitalizing "BookBloom" - A Local Bookstore's Marketing Analytics Journey

- **Introduction:**
- BookBloom is a beloved independent bookstore in a mid-sized city, known for its curated selection and cozy atmosphere. However, they've been facing declining sales and struggling to compete with online retailers and larger chains. The owners realize they need to move beyond traditional marketing methods and embrace data-driven strategies to survive and thrive.

Case Study: Revitalizing "BookBloom" - A Local Bookstore's Marketing Analytics Journey

Problem Statement:

BookBloom faces the following challenges:

- **Declining Foot Traffic:** Fewer customers are visiting the physical store.
- **Limited Online Presence:** Their website is basic, lacking e-commerce functionality and engagement features.
- **Ineffective Marketing Spend:** They rely on sporadic local newspaper ads and flyers, with no clear understanding of their impact.
- **Lack of Customer Insights:** They don't know their customers' preferences or buying habits.
- **Difficulty Measuring Marketing ROI:** They can't track the effectiveness of their marketing efforts.

Case Study: Revitalizing "BookBloom" - A Local Bookstore's Marketing Analytics Journey

Objectives:

- Increase foot traffic by 20% within 6 months.
- Develop a robust online presence and drive online sales.
- Optimize marketing spend based on data-driven insights.
- Enhance customer engagement and loyalty.
- Establish a system to measure and improve marketing ROI.

Case Study: Revitalizing "BookBloom" - A Local Bookstore's Marketing Analytics Journey

Solutions and Examples:

1. Building a Strong Foundation with Marketing Analytics:

•**Solution: Implementing a Point of Sale (POS) System with Data Analytics.**

- Example:** Upgrade to a modern POS system that tracks sales data, customer purchase history, and inventory. This will provide valuable insights into popular genres, customer preferences, and peak shopping hours.

•**Solution: Setting up Google Analytics for Website Tracking.**

- Example:** Implement Google Analytics to track website traffic, bounce rates, time spent on pages, and conversion rates. This will help understand user behavior and identify areas for improvement.

•**Solution: Creating a Customer Relationship Management (CRM) System. Example:**

Implement a CRM to collect customer data (email addresses, purchase history, preferences). This will enable personalized marketing campaigns and targeted promotions.

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2. Optimizing Marketing Spend with Data-Driven Insights:

- Solution: Analyzing Customer Acquisition Cost (CAC) for Different Channels.**
 - Example:** Track the cost of acquiring a customer through different channels (e.g., social media ads, email marketing, local events). Identify the most cost-effective channels and allocate budget accordingly.
- Solution: Attribution Modeling for Multi-Channel Marketing.**
 - Example:** Use attribution models (e.g., last-click, first-click, linear) to understand how different touchpoints contribute to sales. This will help optimize campaigns across multiple channels.
- Solution: A/B Testing for Marketing Campaigns.**
 - Example:** Run A/B tests on email subject lines, social media ad creatives, and website landing pages to determine which versions perform best.

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3. Enhancing Customer Engagement and Loyalty:

- **Solution: Personalized Email Marketing Campaigns.**

- **Example:** Segment customers based on their purchase history and preferences.
- Send targeted emails with book recommendations, event invitations, and exclusive discounts.

- **Solution: Building an Online Community through Social Media.**

- **Example:** Create engaging content on social media (e.g., book reviews, author interviews, behind-the-scenes glimpses). Run contests and polls to encourage interaction and build a community around the bookstore.
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- **Solution: Implementing a Loyalty Program.**

- **Example:** Offer rewards and discounts to loyal customers based on their purchase history. This will encourage repeat business and build customer loyalty.

Case Study: Revitalizing "BookBloom" - A Local Bookstore's Marketing Analytics Journey

4. Measuring and Improving Marketing ROI:

- **Solution: Tracking Key Performance Indicators (KPIs).**

- **Example:** Track KPIs such as website traffic, conversion rates, customer acquisition cost (CAC), customer lifetime value (CLTV), and return on ad spend (ROAS).

- **Solution: Regular Reporting and Analysis.**

- **Example:** Generate weekly or monthly reports to analyze marketing performance and identify areas for improvement. Use data visualization tools to present findings in a clear and concise manner.

- **Solution: Continuous Optimization and Iteration.**

- **Example:** Use data-driven insights to continuously optimize marketing campaigns and strategies. Regularly test new ideas and approaches to improve ROI.

Case Study: Revitalizing "BookBloom" - A Local Bookstore's Marketing Analytics Journey

Expected Outcomes:

- 20% increase in foot traffic within 6 months.
- Development of a thriving e-commerce platform and increased online sales.
- Optimized marketing spend based on data-driven insights.
- Enhanced customer engagement and loyalty.
- Establishment of a system to measure and improve marketing ROI.

Conclusion:

By embracing marketing analytics, BookBloom can transform its business and achieve sustainable growth. By focusing on data-driven decision-making, personalized marketing,

and building a strong online presence, they can attract new customers, retain existing ones, and thrive in the competitive retail landscape.

This case study demonstrates the power of marketing analytics in revitalizing a traditional business and ensuring its long-term success.

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3.1.2 Introduction of data science and analytics concepts

- Data science is a broad term that refers to a collection of related discipline data, focusing on the use of data to create new information and knowledge.
- The ultimate goal is to provide useful insights for better marketing decisions.
- Computers have helped to automate the routine task of number crunching and have freed us to concentrate on more creative and meaningful aspects of data science such as setting up a model and interpreting the results of a data analysis.
- However, they have also contributed to the production of significantly more data by becoming data sources themselves.

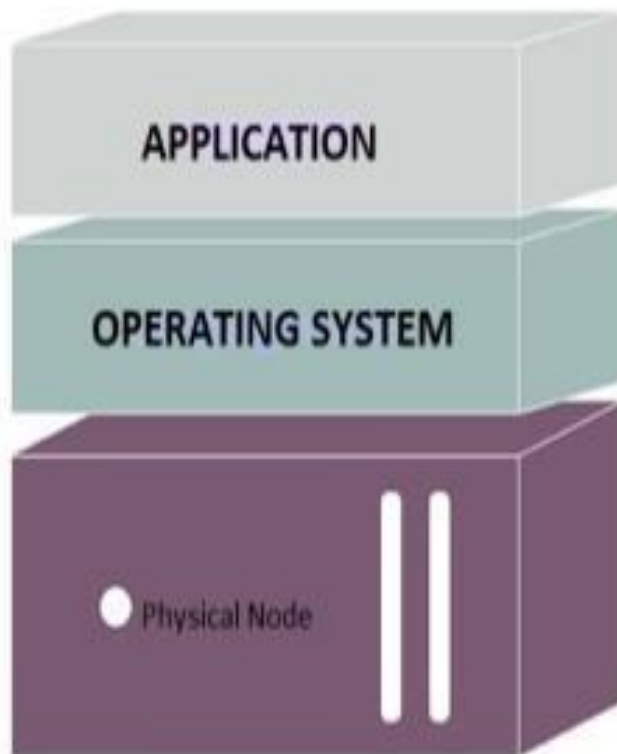
Below are some common trends to study data science:

Virtualization

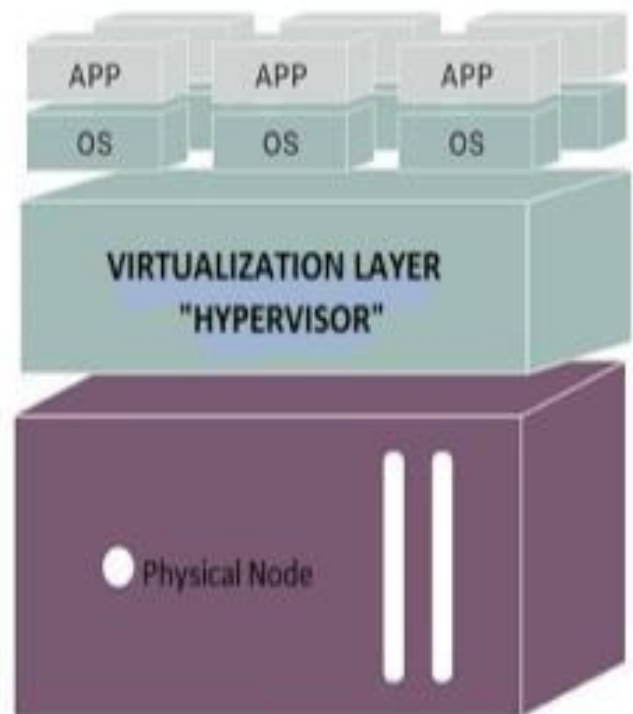
- Virtualization is the creation of a virtual -- rather than actual -- version of something, such as an operating system, a server, a storage device or network resources.
- You probably know a little about virtualization if you have ever divided your hard drive into different partitions. A partition is the logical division of a hard disk drive to create, in effect, two separate hard drives.

The evolution of virtualization

- Operating system virtualization is the use of software to allow a piece of hardware to run multiple operating system images at the same time.
- The technology got its start on mainframes decades ago, allowing administrators to avoid wasting expensive processing power.
- Virtualization platforms like VMware and Microsoft HyperV make this possible.



TRADITIONAL ARCHITECTURE



VIRTUAL ARCHITECTURE

Cloud Computing

- The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer.

Machine Learning

- **Machine learning** is an application of artificial **intelligence** (AI)
- It provides systems the ability to automatically learn and improve from experience without being explicitly programmed.
- It assists in things like sales forecasting, customer segment analysis, insurance claim fraud detection, etc.

3.2.3 Web Traffic nuances

- The amount of traffic and the **details of each visit** are extremely valuable information to a Web-based business.
- The server computer **records every request for a Web page**, allowing its operators to determine which pages get the most attention.
- Web traffic analysis gives businesses concrete, reliable information on the **interests of their customers**.

Evaluation and management of Web Traffic

Sessions:

- When someone visit a Web site, they typically browse through a few different pages before moving on to other sites.
- Web engineers call this visit a session.
- It has a well -defined beginning and an end, though the details vary from one session to the next.

Hits:

- Each Web page may consist of dozens of separate files.
- The server sends each file to your browser, assembles them into a finished piece complete with text and graphics.
- Each file sent constitutes one “hit,” so a single page view results in many hits.
- The more traffic a Web site receives, the more sessions and hits are recorded by servers.

Server Log:

- Every time a Web server processes a file request, the computer makes an entry into a server log, a dedicated file on the server's hard drive.
- The entry contains the Internet address of the user requesting the file, the time and date of the request and the file's name.
- The log may also contain error codes, such as when a user requests a non-existent page.
- The log accumulates many thousands of entries over time, becoming a gold mine of information.
- The site's operators use Web traffic analysis programs to study through the log.
- The analysis reveals the most successful pages on the site, the days and times when users are busiest and from what parts of the world the visits originate.

Capacity:

- Companies can host a modest site on a single server computer and an economical network connection, but a busy site may require dozens of computers and a network that can keep up with the requests.
- Web traffic requires capacity.
- A given site's operators monitor the traffic of the server's processes and use this information to determine when to add more capacity.
- If the operators neglect site capacity, floods in traffic overload the computers and slow the site's responsiveness, resulting in frustrated customers and lost business.

Traffic Analysis:

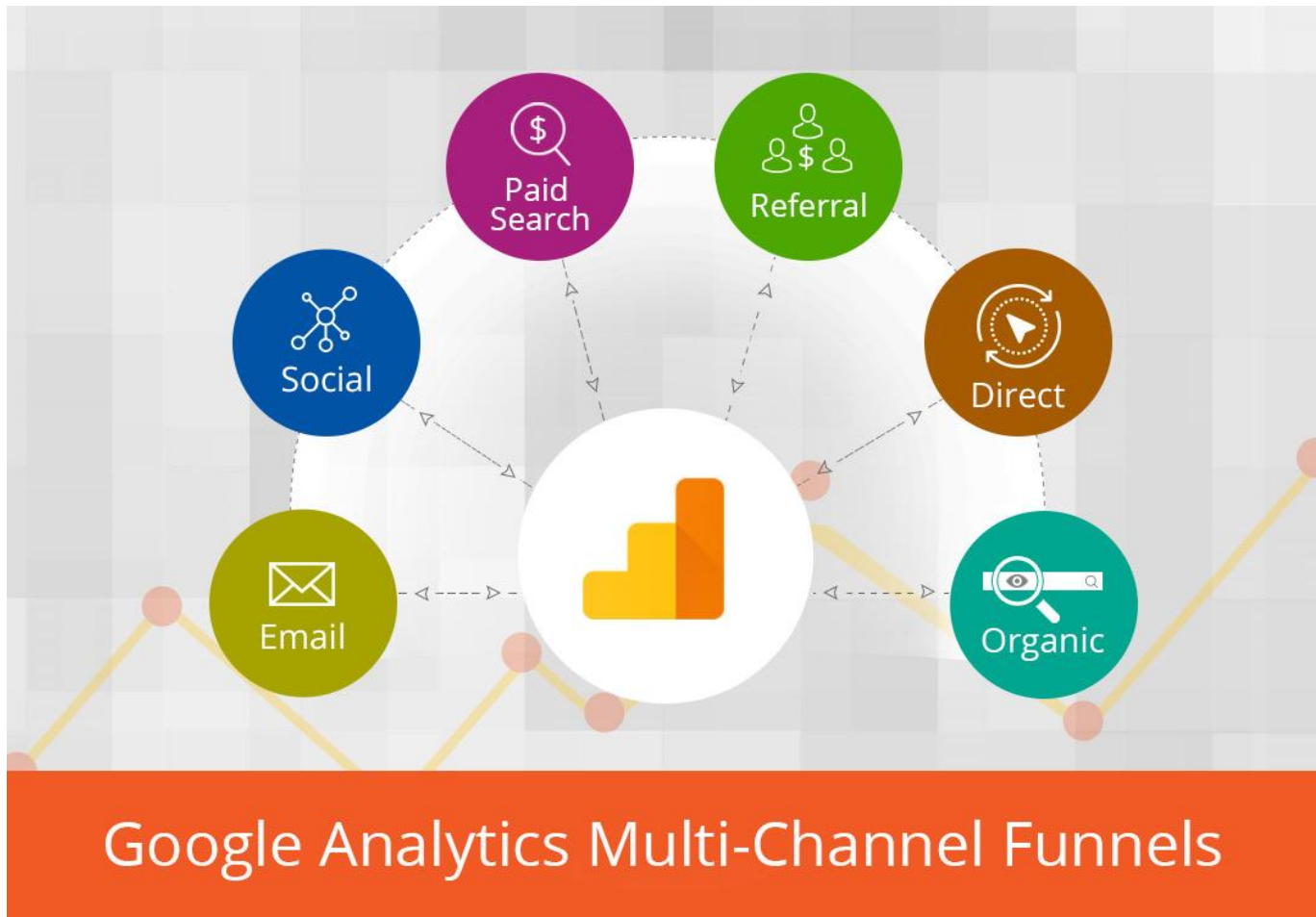
- Determines the effectiveness of business strategies.
- For example, a Web site offering a sale on certain goods can measure the traffic driven by sales.
- The analysis may reveal that the sale pricing wasn't aggressive enough, or it may show customers became interested in other products offered on the site.

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3.2.4. Multichannel Analytics



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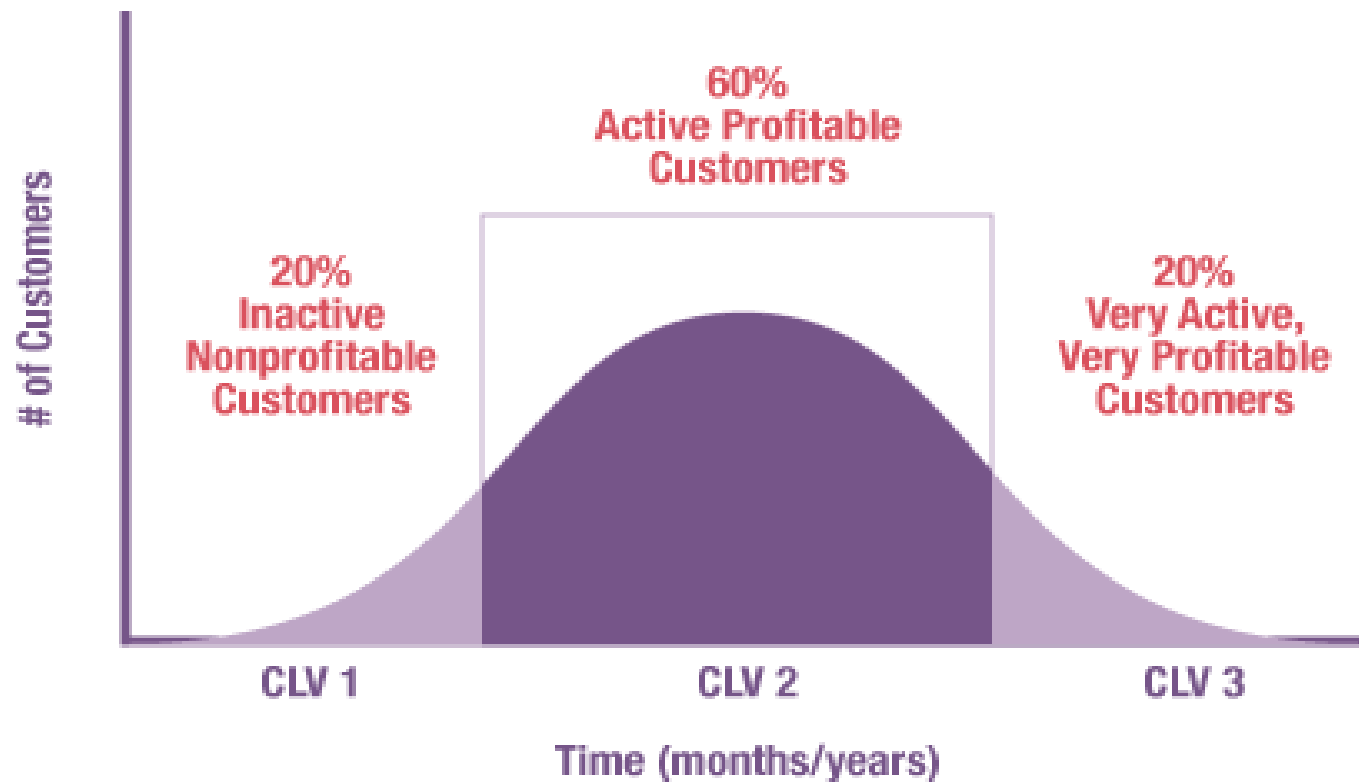
- Multichannel analytics is the practice of getting information from various marketing channels, such as
 - email,
 - traditional,
 - social,
 - offline, etc.,and then combining these pieces of information into one software environment that allows comprehensive reporting and analysis.

- **Goal of multichannel analytics:** is to enable companies to gain valuable business intelligence about their customers and prospects.
- It allows companies
 - to segment customers more efficiently
 - What content and offers are to be sent,
 - when they should be sent and
 - what should be the preferred channels.

3.2.5 Decoding Customer lifetime value CLV and RFM

- It highlights how many customers will have a **long-term relationship** with the business.
- A one-off purchase might show a high cost per acquisition, but if considering repeat purchases, that cost per acquisition creates longer term value for your business.
- It's true that it's much easier to keep an existing customer than it is to win a new one.
- Selling more to an existing customer will also be at a reduced marketing cost.

Customer Lifetime Value



Two major challenges with CLV

- The biggest and most glaring issue with CLV is the use of the word lifetime. What exactly is a lifetime?
- It's not 80 years.
- Do we really need to wait until someone has died before we can calculate the customer lifetime value of our customers? Of course not, and that's why it's important for every business to determine what the specific lifetime is.
- For a business, a customer may transact for an average of two years, so the lifetime would be two years.
- If the customer transacts on average for 15 years, 15 years would be the lifetime in this case. After specifying how long the customer lifetime is, companies have a good estimate of when the customers should be transacting, and more importantly, if they haven't.
- This information will better equip you to know the difference between a customer retained and a customer lost.

- The second challenge with CLV is there is no such thing as an average customer.
- Companies can use averages but it's important to understand that every customer is a person and every person deserves to be treated a little bit better than just an average.

Calculation of CLV

- Most efficient and the easiest model to calculate CLV would be to:
 - take the total profit of a client over the lifetime of buying and subtract all the sales, marketing, and service delivery cost from that client.
- The number you have left is your customer lifetime value number.

Example of CLV calculation

- For example, suppose a new customer brings you **100 dollars in profit on their first purchase.**
- That same customer on average **purchases two more times from you over the next two years.**
- Each time they reorder, they spend approximately 250 dollars with an addition of **100 dollars in profit on each purchase.**
- Total profit per customer during CL = ??? $100+200$
- Therefore, if we do a simple math, that **customer is worth 300 dollars in profit per their lifetime.**

Recency, Frequency and Monetary RFM Calculation Model

- To identify customer lifetime value, a well-known method called recency, frequency and monetary (RFM) model is used to **represent customer behavior characteristics**.
- RFM models have been **used in direct marketing for more than 30 years**.
- These models were developed to target marketing programs at specific customers with the **objective to improve response rates**.
- Prior to these models, companies **typically used demographic profiles of customers** for targeting purposes.
- However, research strongly suggests that past purchases of consumers are better predictors of their future purchase behavior than demographics.

- The basic assumption of using the RFM model is that **future patterns of consumer trading resemble past and current patterns.**
- This study proposes using the following RFM variables
 - **R (Recency)** : the period since the last purchase;
(a lower value corresponds to a higher probability of the customer's making a repeat purchase;)
 - **F (Frequency)**: number of purchases made within a certain period;
(higher frequency indicates greater loyalty;)
 - **M (Monetary)**: the money spent during a certain period;
(a higher value indicates that the company should focus more on that customer)

3.2.6 Deciphering Conversion and Goal Metrics

- Conversion tracking must be present, since that will enable to assign a value to the campaign results.
- It is as easy as adding a piece of PHP code to the website, so the developer can do this in no time.
- Many websites have a standard conversion code installed on all the pages, and with Analytics companies specify which pages to track and count as a conversion.
- Below are some pages which could count as conversion:
 - **a thank-you page**
 - **a receipt page**
 - **a submission-confirm page**
 - **a registration-complete page**

- With Google Analytics, companies can designate specific pages to be tracked,
 - The amount of time a user spent on a site and
 - The number of page views and other various other onsite behaviors.
- They can specify all these metrics as a goal, and then track these goals accurately over time.

How Google Analytics Work

- By understanding how Google Analytics works, companies will be better equipped to identify problems with the setup or understand why certain data anomalies might occur.
- **Setup Google My Business Profile:**<https://youtu.be/chJ--GuLqFw?si=A8jNHt5zwFvk3iue>
- **Google Analytics:**<https://youtu.be/vSeb2vWjRZ8?si=sDVThVXtJuLAKG3A>
- There are four components to how the Google Analytics system works:
 - Data Collection
 - Configuration
 - Data Processing
 - Reporting.

- To track a website, Google Analytics uses a small snippet of **JavaScript code**.
- Developers **place this code on every page of the website** and when a user arrives, the code comes alive and starts **to collect data on how the user engages with the site**.
- Some of the **data comes from the website itself, like the URL of the pages** that the user is viewing.
- Other **data is collected from the user's browser**, like the language the browser is set to, the browser name, and the device and operating system used to access the site.
- The JavaScript can also collect information on **what content is viewed, for how long, and even the referring source** that brought the user to the site.

3.2.7 Implement conjoint analysis & decision tree tactics

- Conjoint analysis involves the measurement of psychological judgments (such as consumer preferences, or acceptance) or perceived similarities or differences between choice alternatives,
- It provides the marketers a greater insight into future predictions and make way through the decision tree.
- The name "Conjoint analysis" implies the study of the joint effects.
- In marketing applications, we study the joint effects of multiple product attributes on product choice.
- Marketing managers are faced with numerous difficult tasks directed at assessing future profitability, sales, and market share for new product entries or modifications of existing products or marketing strategies.

CONJOINT ANALYSIS

Which of these laptops would you choose?

Brand	HP	Dell	Sony
Processor	4GHz	3GHz	2GHz
RAM	1GB	3GB	2GB
Monitor	21-inch	17-inch	15-inch
Price	\$899	\$799	\$699
	☐	☐	☐

- These specific tasks include predicting:
 - The **profitability and/or market share for proposed new product** concepts given the current offering of competitors
 - The **impacts of new competitor products on profits** or market share if we make no change in our competitive position
 - Customer switch rates either from our current products to new products we offer (cannibalism), or from our competitors' products to our new products (draw)
 - The differential response of above items by key market segments purchasing our product
 - Competitive reaction to our strategies of introducing a new product. Specifically, should a new product be introduced, and if so, what is the optimal design configuration for this new product?

3.2.8 Avoiding Common Analytical Pitfalls

- The most common pitfall is jumping to answers too quickly.
- Some of the tools they can use to avoid these pitfalls are - defining your problem well & having a clear problem statement.

- The second pitfall is being unwilling to expand the problem space.
- Sometimes they're going to focus on things that don't matter.
- They'll come up with a recommendation, and even though it's insightful, it won't really change the performance of the organization

- By leveraging the technique of the 80/20 Principle, you can ask yourself, "If I solve this, will it have a unbalanced impact on my organization?" "Or, am I solving for that 80% that drives 20% of the results that don't really matter?"